Symmetries in Physics — Exercise Sheet 1

Groups, examples, and Lagrange's theorem

Variant: Questions only

Exercise 1. Let D_n be the dihedral group with $|D_n| = 2n$, generated by a rotation r and a reflection s. Representing

$$r \mapsto \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{with} \quad \theta = 2\pi/n \quad \text{and} \quad s \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

verify $srs^{-1} = r^{-1}$ directly, and deduce the product rule $(r^a s^\alpha)(r^b s^\beta) = r^{a+(-1)^\alpha b}\, s^{\alpha+\beta}$.

Exercise 2. Show $D_3 \cong S_3$ by an explicit isomorphism between the symmetries of an equilateral triangle and the permutations of its vertices, and verify that it respects one non-commuting product.

Exercise 3. The Klein four-group is

$$V = \{e, a, b, ab\} \quad \text{with} \quad a^2 = b^2 = e \quad \text{and} \quad ab = ba.$$

Show that

(i) V is the symmetry group of a non-square rectangle, and name the four symmetries geometrically.

(ii) $V \ncong C_4$, although both have order four.

(iii) The symmetry group of the planar anisotropic oscillator with potential $W = \frac{1}{2}(k_x x^2 + k_y y^2)$, $k_x \neq k_y$, is again V. Here a symmetry is an orthogonal map of the plane preserving W, hence preserving $H = \frac{1}{2}\left|\vec{p}\right|^2 + W$.

Exercise 4. Classify the groups of order 4: show that any such group is isomorphic either to C_4 or to the Klein four-group V. *Guidance:* by Lagrange every non-identity element has order 2 or 4.

Exercise 5. List all subgroups of the symmetry group D_4 of the square (order 8), give the order and index of each, and verify Lagrange's theorem throughout.

Exercise 6. The groups D_4 and C_8 have order 8. Find the orders of all the elements of each, and conclude that the two groups are not isomorphic.

Exercise 7. Prove the case $p = 2$ of Cauchy's theorem: every group G of even order contains an element of order 2. *Guidance:* pair each element with its inverse.

Exercise 8 ⋆. Find the symmetry group of the FiDi dinosaur.